

DEVELOPER REQUIREMENTS

E-commerce Website

Tracking, Merchant Center & Fulfilment

Project stack: React + Vite frontend + FastAPI backend

Primary objective

Build measurement into the product architecture so every order can be traced from acquisition → onsite behavior → checkout → payment → shipment → delivery/return → realized revenue, while supporting Google Merchant Center and Meta/Google advertising.

Area	Requirement
Frontend	React + Vite, GTM, ecommerce dataLayer, consent-aware browser events
Backend	FastAPI, order/payment/shipping state, server events, webhooks, retries
Google	GA4, Google Ads, Enhanced Conversions, Merchant Center, Search Console
Meta	Meta Pixel + Conversions API with event deduplication
Commerce	Product feed/API, structured data, shipping/returns, payment reconciliation
Attribution	Persist click IDs, UTMs, first/last touch and order-level attribution

1. Architecture decision

Keep React + Vite; choose one crawler-ready rendering path

The frontend remains React + Vite and FastAPI remains the business/API backend. Product/category pages must return crawlable product content, canonical/meta tags and Product/Offer structured data without depending on post-load client JavaScript. Select and document one rendering strategy before implementation.

Rendering options for product/category routes

Option	Implementation	Position
A — React SSR on Vite	Use Vike (formerly vite-plugin-ssr) or an equivalent Vite SSR layer; hydrate the React app after server render.	Preferred when the team wants to keep rendering responsibility in React.
B — FastAPI initial render	FastAPI returns the initial product/category HTML, meta/canonical tags and JSON-LD; React then hydrates/enhances the page.	Valid hybrid option; define one source of truth and prevent markup/hydration mismatch.
C — prerender service	Use a maintained prerender/dynamic-rendering service to return rendered HTML to crawlers while users receive the SPA.	Fallback only; crawler/user primary content and commerce facts must remain equivalent, with cache freshness and equivalence testing to avoid cloaking/stale output.

Moving to Next.js is an optional architectural migration, not a requirement. If the team keeps React + Vite, Option A or B is preferred over a bot-only prerender dependency.

Target architecture:

- React + Vite frontend → FastAPI REST/JSON API → PostgreSQL (system of record).
- Public product/category routes must implement the selected crawler-ready initial HTML strategy and validate the raw server response, not only the hydrated browser DOM.
- GTM Web Container receives browser dataLayer events.
- Server-side tracking layer sends authoritative server events to Meta/Google where appropriate.

- Payment gateway and courier integrations send signed webhooks into FastAPI.
- Background worker/queue handles external API calls, retries and reconciliation without blocking checkout.
- Merchant Center product synchronization reads from the same product/variant source of truth used by the website.

2. Non-negotiable identifier contract

The developer must define stable IDs once and reuse them everywhere. Tracking accuracy will collapse if different systems use different product/order identifiers.

Identifier	Rule	Used by
product_id	Stable parent product ID; never title/slug	DB, catalog, analytics
variant_id / SKU	Stable sellable variant identifier	Merchant, Meta catalog, dataLayer, order_items
item_id	Must map consistently to the sellable SKU/offer ID used in product feeds/catalogs	GA4/Ads/feeds
order_id	Unique immutable commerce order ID	purchase, payment, shipping, refunds
transaction_id	Use order_id or an immutable 1:1 transaction key; never regenerate on refresh	GA4/Google Ads deduplication
event_id	Unique per logical Meta event; same ID in browser Pixel and server CAPI copy	Meta deduplication
customer_id	Internal non-PII identifier; nullable for guest checkout	CRM/LTV/analytics

3. Product & database requirements

Minimum product/variant fields:

- id, parent_product_id, SKU/offer_id, GTIN/barcode when applicable, MPN when applicable, brand
- title, description, category, product_type, tags
- price, sale_price, currency, cost/COGS
- availability/status, stock_quantity, inventory timestamp

- variant attributes: size/color/material etc.; item_group_id for grouped variants
- canonical product URL, image URLs, additional images
- condition, weight/dimensions when relevant to shipping
- Merchant eligibility flags and feed sync timestamps

Minimum order and fulfilment fields:

- order_id, customer_id, created_at, currency, subtotal, discount, tax, shipping, total, COGS/margin when available
- order_status, payment_status, payment_method, gateway_transaction_id
- courier_provider, shipment_id, tracking_number, shipping_cost, fulfilment timestamps
- delivery_status, delivered_at, failed_delivery_reason, return_reason, returned_at
- COD amount, COD collected status/date, COD remitted amount/date
- coupon/promotion IDs
- attribution snapshot stored on the order

4. Acquisition & attribution persistence

Capture on landing, persist through checkout, and copy a snapshot to the final order. Store first-touch and last-touch separately where possible.

Group	Fields to retain
UTM	utm_source, utm_medium, utm_campaign, utm_content, utm_term, utm_id
Google click IDs	gclid, gbraid, wbraid
Meta IDs	fbclid, _fbp, _fbc where available/allowed
TikTok IDs	ttclid, _ttp where the channel is enabled and permitted
Snap IDs	sc_click_id / ScCid, sc_cookie1 where the channel is enabled and permitted
GA context	ga_client_id, ga_session_id; optional internal non-PII user_id where consent allows
Partner/owned	affiliate_id, referral_code, coupon_code
Context	landing_page, referrer, timestamp, consent state
Order snapshot	first_touch_source, last_touch_source plus the applicable IDs above at conversion

Important

Do not depend on cookies alone. When the shopper provides an email/phone or creates an order, associate the allowed attribution data with the server-side cart/order record so the history survives browser loss where legally permitted.

Use this attribution object as a channel-neutral schema. New channel adapters must extend the same order snapshot rather than inventing a separate source of truth. Enable Google Ads auto-tagging and retain GCLID/GBRAID/WBRAID whenever available.

5. Ecommerce dataLayer contract

Create one reusable tracking service in the frontend. Components should call that service; do not hard-code Meta/GA tags throughout UI components. Follow GA4 recommended ecommerce event names and parameters.

Event	Trigger	Core data
page_view	Every route/view, including SPA navigation	page_location, page_title
view_item_list	Category/search/recommendation list rendered	item_list_id/name, items[]
select_item	Product selected from list	item_list_id/name, items[]
view_item	Product/variant detail viewed	currency, value, items[]
view_promotion	Promotion rendered	promotion_id/name, creative_name/slot
select_promotion	Promotion clicked	promotion fields
search	Internal search submitted	search_term
add_to_wishlist	Wishlist add	currency, value, items[]
add_to_cart	Successful cart add	currency, value, items[]
remove_from_cart	Successful removal	currency, value, items[]
view_cart	Cart viewed	currency, value, items[]
begin_checkout	Checkout actually begins	currency, value, coupon, items[]

Event	Trigger	Core data
add_shipping_info	Shipping selection confirmed	shipping_tier, currency, value, items[]
add_payment_info	Payment method confirmed	payment_type, currency, value, items[]
purchase	Order accepted/confirmed exactly once	transaction_id, value, tax, shipping, currency, coupon, items[]
refund	Full/partial refund confirmed	transaction_id, value/currency and affected items[]
sign_up / login	Account actions	method

Each item object should support: item_id, item_name, affiliation (if applicable), coupon, discount, index, item_brand, item_category through item_category5 as relevant, item_list_id, item_list_name, item_variant, location_id (if relevant), price and quantity.

```

window.dataLayer.push({
  event: "purchase",
  ecommerce: {
    transaction_id: "ORD-100245", currency: "EGP", value: 2199.00,
    tax: 0, shipping: 80.00, coupon: "WELCOME10",
    items: [{ item_id: "SKU-BLK-42", item_name: "Example Product",
              item_brand: "Brand", item_variant: "Black / 42", price: 2119.00, quantity: 1 }]
  },
  tracking: {
    logical_event_id: "purchase_ORD-100245",
    meta_event_id: "meta_purchase_ORD-100245",
    tiktok_event_id: "tt_purchase_ORD-100245",
    snap_event_id: "snap_purchase_ORD-100245"
  }
});

```

Destination event IDs must be generated once per logical event and reused by that destination's browser/server copies. Do not reuse one platform-specific identifier namespace as another platform's deduplication key. The immutable order_id remains the shared commerce identity.

6. GTM, GA4 & Google Ads

- Install GTM once at the application shell/document level and make it consent-aware.
- Use separate staging/test and production GTM Web + Server containers (or an equivalently isolated release setup) so QA events cannot contaminate production data. Keep environment-specific IDs/configuration outside shared client logic.
- Configure GA4 ecommerce from the dataLayer rather than duplicating event logic in tags.

- Track virtual page views correctly for client-side navigations.
- Deploy server-side GTM on a first-party custom endpoint such as `metrics.example.com`; same-origin serving/path is preferred when practical. This improves first-party context and cookie durability, but must not be described as guaranteed ad-blocker bypass.
- Use GA4 Measurement Protocol for authoritative server/offline events such as delivered/returned outcomes when GA4 reporting needs them. Preserve the required client/session identifiers and `transaction_id` so server events can be associated and deduplicated correctly.
- Google Ads purchase conversion must use dynamic value/currency and stable `transaction_id`.
- Implement Enhanced Conversions for Web using consented first-party customer data where permitted.
- Enable Google Ads auto-tagging. Preserve `gclid/gbraid/wbraid` for attribution and later server/offline conversion workflows; when supported by the chosen offline workflow, retain permitted session attributes/matching data as a fallback rather than replacing click-ID capture.
- For COD, upload an authoritative Delivered conversion to Google Ads after courier confirmation. For new 2026 integrations use Google Ads Data Manager API for offline/enhanced-lead conversion ingestion; persist `GCLID/GBRAID/WBRAID` plus `order_id`, event time/value and consented first-party identifiers where applicable. Keep Delivered separate from website Purchase to avoid double-counting purchase revenue.
- Define conversion-goal governance before campaigns launch: Purchase, Delivered, AddToCart and BeginCheckout must not all become bidding conversions by accident. Mark actions Primary or Secondary deliberately and document any campaign-specific/custom goal.
- Recommended default: website Purchase is Primary when campaigns optimize to placed/paid orders. For COD-only campaigns, Delivered may become the Primary optimization action only when the business intentionally optimizes to successful delivery and has enough stable data; in that setup avoid simultaneously optimizing to Purchase and Delivered as equivalent revenue outcomes. Micro-conversions remain Secondary unless a documented test says otherwise.
- If GA4-imported Purchase and a native Google Ads Purchase both exist, choose one bidding source of truth. Keep the duplicate action Secondary/diagnostic so reporting and Smart Bidding are not inflated.
- Upload Google Ads conversion adjustments for cancellations, returns and corrected values: map the authoritative backend order lifecycle to RETRACTION or RESTATEMENT using the original transaction/order identity and conversion action.
- Where campaign/reporting needs product economics, implement Conversions with Cart Data: send the same Merchant-compatible product IDs plus item quantity/value; COGS may be supplied where supported and trustworthy.
- Expose a server-derived `new_customer` flag for Google new-customer acquisition reporting/bidding where used; do not infer it only from a browser cookie. Customer-lifetime-value features are optional and require a documented business definition.

- Validate in GTM Preview/Tag Assistant, GA4 DebugView and Google Ads diagnostics before launch.

7. Meta Pixel + Conversions API

Browser Pixel alone is not sufficient for robust measurement. Implement browser + server signals for material commerce events, using the same logical event_id for Meta deduplication.

Browser event	Server counterpart	Key rule
ViewContent	Optional/selected high-value pages	Use product content_ids matching Meta catalog IDs
AddToCart	Recommended	Same value/currency/content IDs
InitiateCheckout	Recommended	Consistent value/currency
Purchase	Required	Server source of truth; same event_id as browser copy
Custom: OrderDelivered	Recommended for COD quality optimization/reporting	Never send a second Purchase for the same order simply because it delivered
Custom: OrderReturned/Cancelled	Optional operational quality signal	Keep separate from Purchase; reconcile business revenue internally

- CAPI payload should include event_name, event_time, event_id, event_source_url/action_source, currency/value and product IDs.
- Use allowed customer matching parameters such as _fbp, _fbclid, external_id, hashed email/phone and IP/user agent where consent, policy and applicable law allow.
- Meta content_ids must map to the same canonical sellable product/variant IDs used by the Meta catalog.
- Never expose Meta access tokens in React. Store secrets only server-side.
- Log CAPI response/status without logging raw sensitive customer data.
- Verify Event Match Quality and browser/server deduplication in Meta Events Manager.

Optional TikTok & Snapchat server signals

If these ad channels are activated, extend the same browser + server measurement contract instead of creating unrelated tracking logic:

- TikTok: Pixel + Events API may mirror the same conversion; send the same event_id on both copies for deduplication. Retain ttclid and _ttp, and use allowed matching signals such as external_id, hashed email/phone and IP/user agent when permitted. content_id must match the TikTok Catalog when catalog ads are used.
- Snapchat: Pixel + Conversions API must use matching identifiers for duplicated events (Pixel client_dedup_id ↔ CAPI event_id), plus the same event name/Pixel context. Retain sc_click_id/ScCid and sc_cookie1 plus allowed external_id/email/phone matching signals where permitted; product IDs must match the Snap Catalog when used.
- Reuse the canonical order_id/product IDs, value, currency, timestamps, consent state and server-side secrets. Never create separate order attribution logic per ad platform.

Optional channel extension contract

If additional paid/owned channels are activated later, add a thin adapter to the same event and attribution contracts. Examples: Microsoft Ads (UET/MSCLKID plus offline conversions where used), Pinterest (Tag + Conversions API), and partner/influencer/email/SMS/WhatsApp acquisition using UTMs plus affiliate/referral codes. Each enabled channel must document browser event mapping, server event mapping if supported, deduplication key, click/session identifiers, catalog product-ID mapping, consent rules and its single conversion source of truth.

8. Google Merchant Center

Single source of truth

Website price/availability, structured data and Merchant product data must all be generated from the same product/variant record. Never maintain three manually independent copies.

Developer deliverables:

- Product/Offer JSON-LD on every purchasable product/variant page, present in the initial HTML returned to the crawler via pre-render/SSR/static generation. Price, currency and availability must match what the shopper sees.
- For variants, implement stable variant URLs/identifiers and ProductGroup/variant structured data where appropriate.
- Product data source using Merchant API v1 or a scheduled feed. Do not build a new integration against the retired Content API for Shopping.
- Feed/API fields as applicable: offer ID, title, description, link, image_link, additional images, availability, price/sale price, brand, GTIN/MPN, condition, item_group_id, color/size/gender/age group for applicable categories, shipping/product weight attributes where required.

- Automatic inventory/price synchronization after catalog changes, plus scheduled full reconciliation.
- Public image/product URLs accessible to Googlebot; no auth wall or robots/noindex on eligible product pages.
- XML sitemap(s), canonical URLs, robots.txt, 404 handling and Search Console setup.
- If the storefront serves multiple countries/languages, implement stable locale/country URLs and hreflang annotations. Currency shown on each landing page must match that market's Merchant product data; keep country/feed labels and shipping settings explicit rather than silently switching currency after landing.
- Shipping and return policies represented consistently on the website and configured in Merchant Center; structured policy markup where applicable.
- Merchant diagnostics monitoring: rejected products, price mismatch, availability mismatch, broken images and feed/API errors.

Merchant storefront compliance pages to expose clearly: Contact/About or business identity, Shipping Policy, Returns/Refund Policy, Privacy Policy, Terms & Conditions, supported payment methods and checkout information. Exact policy/eligibility depends on target country/product category and must be reviewed before launch.

Cross-platform product catalog contract

The product/variant source of truth must be able to publish channel-specific catalog/feed adapters without duplicating catalog business data. Google Merchant Center is required; Meta Catalog, TikTok Catalog and Snap Catalog should be enabled when those advertising channels use dynamic/catalog commerce formats.

- The canonical sellable SKU/offer ID must remain consistent across the database, website dataLayer, GA4/Google Ads and every enabled ad catalog. Channel adapters may transform field names, but must not invent a different product identity.
- Each enabled adapter must map title/description, product URL, image, availability, price/sale price, currency, brand/GTIN where applicable, variant grouping and market eligibility from the same source record.
- Catalog sync must support incremental updates after inventory/price changes plus scheduled full reconciliation, with per-channel diagnostics for rejected/stale/mismatched items.
- A catalog product-ID contract is part of launch QA: a product clicked/viewed/purchased must resolve to the same sellable item used by that channel's catalog.

8A. Technical SEO & organic discoverability

SEO is an application requirement, not a GTM tag

Critical SEO output must be produced by the storefront/rendering layer. For every indexable product, category and SEO landing page, the raw initial HTML should contain the main content, title, meta description, canonical/hreflang where applicable and the relevant structured data.

Indexability & URL contract

- Use stable, descriptive, crawlable URLs. Do not use # fragments as the identity of indexable product variants. If a slug changes, keep redirect history and issue a real server-side 301 to the new canonical URL.
- Define an indexability matrix: products, categories and intentional content/SEO landing pages are normally indexable; cart, checkout, account/login, order tracking, internal search and thin/duplicate sort/filter combinations are normally noindex unless an explicit SEO case exists.
- Do not block a URL in robots.txt when relying on its robots noindex directive; the crawler needs access to read that directive. Use robots.txt primarily to manage crawling, not as a substitute for de-indexing.
- Each clean indexable URL should normally emit a consistent self-canonical. Duplicate sort/filter/tracking-parameter URLs need one documented canonical/indexing policy; never emit contradictory canonical and hreflang signals.
- Faceted navigation requires an allowlist: create crawlable/indexable filter landing pages only for combinations with genuine search demand and unique useful content; control the remaining combinations to prevent effectively infinite crawl spaces.
- Category → subcategory → product, breadcrumbs and pagination must use real crawlable <a href> links. Critical discovery cannot depend on button clicks or client-only actions. Infinite scroll may enhance UX, but each paginated chunk still needs a unique crawlable URL/link path.

On-page, structured data & international SEO

- Provide unique server-rendered <title>, meta description and meaningful H1 for indexable templates. CMS/admin fields should include slug, seo_title, meta_description, index/noindex, optional canonical override, Open Graph title/description/image, image alt text and redirect history.
- Structured data must match visible content and the commerce source of truth. Use Product/ProductGroup/Offer on product pages and BreadcrumbList on navigational hierarchies; add Organization/site-level markup only where applicable.
- For multi-country/language storefronts, use stable locale URLs and reciprocal hreflang annotations including the page itself; x-default may identify a neutral/default selector. Avoid forcing every crawler/user to another country solely by IP. Currency, availability and Merchant data must match the landing market.

- Images need stable crawlable URLs, descriptive alt text, explicit dimensions/responsive delivery and performance-appropriate formats. Keep the primary product image discoverable in initial markup.

Sitemaps, status codes, performance & monitoring

- Generate XML sitemap index/files automatically from the database. Include only canonical, indexable, HTTP-200 URLs; split products/categories/content when useful; keep <lastmod> accurate and submit the sitemap in Google Search Console. Reference it from robots.txt where appropriate.
- Return real HTTP status codes: 301 for permanent migrations, 404/410 for removed/not-found content as appropriate, and 5xx for genuine server failures. Do not return a branded error page with HTTP 200 (soft-404 behavior).
- Performance acceptance target at the 75th percentile: LCP ≤ 2.5 s, INP < 200 ms and CLS < 0.1 . Optimize the LCP asset, reserve image/layout dimensions, keep JavaScript/hydration work controlled and lazy-load non-critical below-the-fold media.
- Configure Google Search Console and monitor Page indexing, sitemap errors, Core Web Vitals, rich-result/Merchant reports and manual actions. Operational monitoring should also flag broken canonicals/hreflang, orphan products, 404/5xx spikes and storefront-vs-structured-data/feed mismatches.

SEO acceptance check

For representative product/category/locale URLs, QA must inspect the raw HTTP HTML (before hydration), HTTP status, canonical/hreflang, robots directives, crawlable internal links, structured data, sitemap membership and performance. A browser DOM that becomes correct only after JavaScript is not sufficient for this project contract.

9. Checkout & payment tracking

Never treat a button click or thank-you page load as the sole payment truth. The backend owns order/payment state and consumes gateway webhooks.

Payment redirects & attribution

If checkout redirects through another domain, preserve attribution at the server/order level first. Configure GA4 cross-domain measurement only for domains where your tagging/linker setup can operate; for external payment processors, also configure unwanted referrals where appropriate so the gateway does not replace the original acquisition source. Always reconcile the return URL with the gateway webhook and immutable order_id.

State/event	Source of truth	Tracking action
order_created	FastAPI DB	Create immutable order ID; snapshot attribution
payment_initiated	Backend/gateway	Operational event; no revenue conversion yet
payment_succeeded	Signed gateway webhook	Update payment_status; reconcile gateway transaction
payment_failed	Gateway/backend	Track failure reason category; do not expose sensitive details
purchase	Order confirmation policy	Fire once; idempotent by transaction_id/order_id
refund	Gateway/admin confirmation	GA4 refund; update realized revenue

Idempotency

Every payment/shipping webhook handler must be idempotent. The same provider webhook can arrive more than once; duplicate delivery must not create duplicate orders, conversions or refunds.

10. Bosta / courier integration

Build a courier abstraction rather than Bosta-specific business logic throughout the codebase. Bosta can be the first adapter; another courier can be added later without changing the order model or tracking contract.

```
CourierProvider interface
  create_shipment(order) -> shipment
  cancel_shipment(shipment_id)
  get_tracking(shipment_id) -> provider_status
  handle_webhook(payload, headers) -> normalized_event
  normalize_status(provider_status) -> internal_status
```

Normalize provider-specific statuses into internal statuses:

- shipment_created
- ready_for_pickup / picked_up
- in_transit
- out_for_delivery

- delivered
- delivery_failed / postponed
- return_initiated
- returned / return_received
- cancelled
- COD_collected / COD_remitted when the provider exposes these states

Courier webhook requirements:

- Dedicated endpoint, e.g. POST /webhooks/shipping/{provider}.
- Verify provider signature/secret when the provider supports it; reject invalid payloads.
- Store provider event ID/raw-status reference and processed timestamp for deduplication/audit.
- Respond quickly, queue processing, and retry transient downstream failures.
- If the selected courier does not offer the required webhook, implement scheduled status polling + reconciliation as fallback.
- Do not assume Bosta endpoint names/status strings in business code; confirm against the account's current API documentation during implementation.

11. COD and realized-revenue measurement

For Egypt/COD, the marketing dashboard must distinguish an order from a delivered, paid order.

Minimum funnel:

Metric	Definition
Orders	Accepted website orders
Shipped	Shipment handed to courier
Delivered	Courier confirms delivery
Delivery rate	$\text{Delivered} \div \text{Shipped}$ (or Orders; label denominator explicitly)
Returned/failed	Shipment not realized
Gross ordered revenue	Value at order creation
Delivered revenue	Value of delivered orders
Net realized revenue	Delivered revenue minus refunds/returns and defined cost adjustments
CAC per delivered order	$\text{Ad spend} \div \text{delivered orders}$

Advertising optimization

Keep 'Purchase' and 'Order Delivered' as distinct conversion concepts. This allows ads/reporting to see both checkout volume and actual fulfilment quality without duplicating the same purchase revenue.

11A. Profit, customer value & BI-ready measurement**Measure business outcome, not revenue alone**

The backend must retain the cost and customer dimensions needed to move from ROAS to realized profit. Advertising platforms remain activation/reporting destinations; the database is the financial measurement source of truth.

Order-level profit ledger

- Persist product COGS at order-item level as a historical snapshot, plus discount/promotion cost, shipping/courier cost, COD fee, payment-gateway fee and any defined return/refund cost. Do not recalculate old orders from today's product cost.
- Define Gross ordered revenue, Delivered revenue and Net realized revenue separately from contribution profit. A recommended contribution-profit definition is Net realized revenue minus COGS minus attributable fulfilment/payment/return costs; document the exact business formula.
- Expose ROAS and profit-oriented metrics side by side: ad spend, ordered revenue, delivered revenue, net realized revenue, contribution profit, CAC per delivered customer and POAS/profit-to-ad-spend when spend data is available.
- All cost/value corrections must be auditable by order_id and timestamp so dashboards can be reconciled after returns, refunds, courier remittance or fee adjustments.

Customer & lifetime-value layer

- Maintain a stable internal customer_id that can associate permitted orders across sessions without exposing PII to the browser. Guest identity merging must use explicit, auditable rules.
- Derive new_customer vs returning_customer from authoritative order history, then expose first_order_at, orders_count, delivered_orders_count, lifetime_net_revenue and lifetime_contribution_profit where available.

- Support cohort/LTV reporting and CAC:LTV analysis by acquisition source/campaign while retaining the original first-touch acquisition snapshot. Do not overwrite first acquisition when a returning customer comes through a new campaign.
- Google/Meta or other platform LTV/new-customer features may consume this layer where enabled, but platform-specific definitions must not become the customer master record.

Warehouse / BI readiness

- Design analytics-ready entities or views for orders, order_items, customers, attribution, payments, shipments/status history, refunds/returns, marketing costs and catalog products. Keep immutable IDs and UTC/event timestamps plus business timezone context.
- Provide a documented export/ELT path to a warehouse/BI destination such as BigQuery and reporting tools such as Looker Studio when required; the launch can begin on PostgreSQL, but schemas and keys must not block later warehouse replication.
- Create reconciliation views/jobs that can compare Ads spend/conversions → website orders → delivered orders → returns/refunds → net realized revenue → contribution profit by date/channel/campaign.
- Marketing dashboards must label attribution model and revenue stage explicitly; do not compare platform-attributed conversion totals to backend realized revenue as though they were the same metric.

12. Consent, privacy & first-party data

- Implement a consent banner/CMP appropriate to the markets served.
- Implement Google Consent Mode v2 with default state before measurement tags and update state immediately after user choice.
- Consent defaults must be region-aware where the approved legal/CMP design requires different behavior by market (for example, an EU/EEA visitor may require a different default than a visitor in another region). Resolve region before optional tags where practical, default conservatively when region is uncertain, and keep the policy/configuration documented rather than hard-coding one global consent state in React.
- GTM/Pixel/CAPI behavior must honor the site's consent policy and applicable law. For Egypt, explicitly assess compliance with Personal Data Protection Law No. 151 of 2020 and its 2025 Executive Regulations (Ministerial Decree No. 816/2025), including consent/legal basis, notices, data-subject rights, retention/security, processor contracts and cross-border transfers where applicable.
- If the business is established in the EU/EEA or the processing is in scope because the site offers goods/services to, or monitors the behavior of, people in the EU/EEA, add a GDPR compliance

review (lawful basis/consent where required, transparency, rights, processor terms, retention, security and international-transfer safeguards). Do not assume GDPR applies merely because the website is globally accessible.

- Do not send raw sensitive customer data to dataLayer. Keep PII out of browser debugging surfaces.
- Hash/normalize customer data only where required by the destination specification; keep secrets and raw transformations server-side.
- Define retention/deletion rules for customer and tracking data.

13. Backend reliability & security requirements

- PostgreSQL transactions for critical order/payment/inventory updates.
- Background job queue for Merchant sync, CAPI, courier/gateway work and retryable webhooks.
- Idempotency keys for checkout/order creation and external webhook processing.
- Exponential retry for transient third-party failures plus a dead-letter/failed-job review path.
- Webhook signature verification, rate limiting and replay protection where supported.
- Secrets in server-side environment/secret manager only; never exposed through Vite client bundles or VITE_* public environment variables.
- Structured logs using IDs (order_id, shipment_id, event_id) while minimizing PII.
- Admin audit trail for manual order/payment/shipping/refund status changes.
- Monitoring/alerts for failed Merchant/catalog sync, webhook failures, CAPI/server-event API failures and checkout errors.
- Tracking-health alerts for abnormal drops/zeros or duplication spikes in Purchase/server events, failed destination responses, stale catalog/feed sync, price/availability mismatches and missing courier status updates. Alerts must include environment, destination/provider and a traceable order/event reference where applicable.
- Scheduled reconciliation must surface gaps between order/payment/courier truth and analytics/ad-platform delivery, rather than relying only on platform dashboards to reveal tracking failures.
- Backups and database migration process before production launch.

14. Recommended API/endpoints

Endpoint / capability	Purpose
GET /products, /products/{slug}	Catalog/product source for frontend
POST/PATCH /cart	Server-associated cart where business needs it
POST /checkout / orders	Idempotent order creation

Endpoint / capability	Purpose
POST /webhooks/payment/{provider}	Authoritative payment status updates
POST /webhooks/shipping/{provider}	Authoritative/frequent fulfilment updates
POST /tracking/server-event (internal)	Queue controlled server conversion events; never public arbitrary forwarding
Merchant sync worker	Push/reconcile product data via Merchant API v1 or configured feed
Reconciliation jobs	Payment, courier, Merchant and tracking gap detection

15. QA & launch acceptance tests

- ☐ Product view produces exactly one correct view_item for the selected variant.
- ☐ List impressions/selections carry correct item_list_id/name and item IDs.
- ☐ Cart add/remove quantities and values are correct after rapid repeated clicks.
- ☐ Checkout events fire on actual state transition, not merely UI button click.
- ☐ Purchase fires once after refresh/back navigation and has stable transaction_id.
- ☐ Destination-specific event IDs (Meta/TikTok/Snap where enabled) are generated once and reused by that platform's browser/server copies without cross-platform namespace collisions.
- ☐ Google Ads has one documented Primary bidding conversion per intended campaign goal; duplicate/native-vs-GA4 actions and micro-conversions are Secondary unless explicitly tested.
- ☐ Google Ads conversion adjustments correctly retract cancelled/returned orders and restate corrected conversion values without creating a new order conversion.
- ☐ Google Ads cart-data product IDs/quantities match Merchant IDs when Conversions with Cart Data is enabled.
- ☐ Meta browser + CAPI copies deduplicate by the same event_id.
- ☐ If TikTok/Snap server APIs are enabled, browser/server copies deduplicate with the platform-required matching IDs.
- ☐ UTMs/click IDs survive landing → navigation → checkout and are copied to the order record.

- ☐ A payment-gateway redirect does not overwrite the original source/medium or detach the order from its attribution snapshot.
- ☐ Guest checkout and logged-in checkout both track correctly.
- ☐ COD and online payment flows produce correct statuses.
- ☐ Duplicate gateway/courier webhooks do not duplicate conversions or state transitions.
- ☐ Partial/full refund updates analytics and order ledger correctly.
- ☐ Courier delivery and return states map correctly into normalized internal statuses.
- ☐ Merchant product ID equals tracking/catalog ID contract; price/availability match page and feed/API.
- ☐ Structured data is visible in initial server-returned HTML and passes Google Rich Results testing where eligible.
- ☐ Representative product/category URLs expose correct raw-HTML title, H1, canonical, robots, hreflang and crawlable links before hydration.
- ☐ Sitemaps contain only intended canonical/indexable 200 URLs; filter/search/cart/checkout indexability follows the approved matrix.
- ☐ Production field performance is monitored against $LCP \leq 2.5$ s, $INP < 200$ ms and $CLS < 0.1$ at the 75th percentile.
- ☐ For rendering Option A or B, hydration does not visibly replace/flash critical product content, move major layout elements or delay the LCP element. Compare server HTML → first paint → hydrated state and fail QA on hydration-induced layout shift/content mismatch; CWV targets remain $LCP \leq 2.5$ s, $INP < 200$ ms and $CLS < 0.1$ at the 75th percentile.
- ☐ If Option C prerendering is selected, automated/manual equivalence tests compare crawler-rendered and user-rendered primary content, price, availability, links, canonical/meta and structured data. Cache invalidation/freshness is tested after product, price, stock and SEO changes so bots are not served stale or materially different content.
- ☐ Staging/test GTM and analytics traffic is isolated from production containers/properties/streams.
- ☐ A courier-confirmed COD delivery can be sent to GA4/server reporting and Google Ads offline conversion ingestion without creating a second website Purchase.
- ☐ Consent denied/granted states change tag behavior as designed; Consent Mode verified in Tag Assistant.
- ☐ Consent defaults are tested for each supported regional policy/configuration, including conservative fallback behavior when region cannot be resolved.

- ☐ Enabled Google/Meta/TikTok/Snap catalogs resolve the same sellable SKU/offer ID used by dataLayer/order_items, and price/availability reconciliation passes.
- ☐ A delivered/refunded/returned test order reconciles from gross order value through fees/COGS/return costs to net realized revenue and contribution profit using the documented formula.
- ☐ A new customer and a returning customer produce correct customer_id, new/returning classification, first-touch preservation and LTV aggregates.
- ☐ BI/reconciliation output can trace Ads → order → payment → shipment → delivery/return → net revenue → contribution profit using stable IDs.
- ☐ Tracking-health monitoring is tested by simulating at least one failed server-event destination, stale catalog sync and missing/failed webhook; alerts are actionable and do not expose raw PII.
- ☐ 404, payment failure and checkout technical errors are observable in error monitoring.

16. Definition of Done

The project is NOT tracking-complete merely because GA4 and Pixel receive events.

Done means a test order can be followed end-to-end from original acquisition identifiers through product/cart behavior, checkout, payment, shipment, delivered/returned state and realized revenue, with Merchant data consistent with the storefront and with no duplicate conversions.

- ☐ DataLayer specification documented and implemented, including destination-specific event-ID deduplication contract.
- ☐ Production and staging GTM/analytics configurations isolated and documented.
- ☐ First-party server-side GTM endpoint configured and validated.
- ☐ GA4 ecommerce + Measurement Protocol server events validated where used, with ga_client_id/ga_session_id association where permitted.
- ☐ Google Ads dynamic Purchase + Enhanced Conversions + COD Delivered offline conversion workflow validated.
- ☐ Google Ads Primary/Secondary bidding-goal governance, duplicate-source rule and cancellation/return value adjustments validated.
- ☐ Meta Pixel + CAPI with deduplication validated.
- ☐ TikTok/Snap browser + server deduplication validated when those channels are enabled.
- ☐ Universal attribution snapshot stores applicable UTM, Google, Meta, TikTok, Snap, GA session and partner/referral identifiers under consent rules.

- ☐ Product/variant IDs consistent across DB, GA4 and every enabled ad catalog/feed including Merchant.
- ☐ Google Merchant plus enabled Meta/TikTok/Snap catalog adapters synchronize/reconcile price, availability and sellable IDs from the same catalog source of truth.
- ☐ Merchant API/feed + crawler-ready Product/Offer structured data in initial HTML validated.
- ☐ Technical SEO acceptance passed for raw HTML, canonicals/hreflang, indexability, crawlable internal links, sitemaps, status codes, Search Console and hydration/Core Web Vitals behavior.
- ☐ Option C, if used, passes crawler/user content-equivalence and prerender cache-freshness QA.
- ☐ Payment redirect attribution/cross-domain or unwanted-referral behavior tested.
- ☐ Payment and courier integrations/webhooks tested with retries/idempotency.
- ☐ Order-level attribution persisted in database.
- ☐ Profit ledger supports COGS, discounts, fulfilment/payment/COD fees and return/refund costs with an auditable net-revenue/contribution-profit formula.
- ☐ Customer layer supports new/returning classification, repeat purchase and LTV/CAC:LTV analysis while preserving first acquisition.
- ☐ BI-ready schemas/reconciliation can trace ad spend and conversions through order, delivery/return, net revenue and contribution profit.
- ☐ COD delivery/return/revenue dashboard data available.
- ☐ Consent Mode/privacy behavior tested, including region-aware defaults and Egypt PDPL/GDPR applicability review.
- ☐ Tracking-health monitoring, failure alerts and reconciliation jobs enabled and test-triggered before launch.
- ☐ Developer provides a tracking map, SEO/indexability matrix, profit metric dictionary, catalog ID mapping, data/BI schema, environment-variable list, webhook setup instructions and QA evidence at handover.

17. Suggested implementation phases

Phase	Deliverable
1 — Data model	IDs, product/variant/order/payment/shipping/attribution, profit ledger and customer/LTV schemas

Phase	Deliverable
2 — Storefront & SEO	React + Vite pages, crawlable initial HTML, URL/indexability contract, structured data, sitemaps, CWV/Search Console
3 — Browser measurement	dataLayer + isolated GTM environments/containers + GA4 + ad pixels + consent
4 — Commerce integrations	Payment gateway + courier adapter/webhooks
5 — Server measurement	first-party sGTM, Meta CAPI, GA4 MP, Google offline Delivered/adjustments; TikTok/Snap APIs if enabled
6 — Catalogs & BI	Merchant + enabled Meta/TikTok/Snap catalogs; BI/export views and reconciliation
7 — QA	End-to-end tracking/profit/SEO/CWV/equivalence tests, alert simulation, reconciliation and documentation

18. Authoritative implementation references

GA4 ecommerce measurement: <https://developers.google.com/analytics/devguides/collection/ga4/ecommerce>

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Google Tag Manager data layer: <https://developers.google.com/tag-platform/tag-manager/datalayer>

Server-side GTM custom domain: <https://developers.google.com/tag-platform/tag-manager/server-side/custom-domain>

GA4 Measurement Protocol: <https://developers.google.com/analytics/devguides/collection/protocol/ga4>

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Google Consent Mode: <https://developers.google.com/tag-platform/security/guides/consent>

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Google Ads conversion adjustments: <https://support.google.com/google-ads/answer/7686447>

Google Ads API — upload conversion adjustments: <https://developers.google.com/google-ads/api/docs/conversions/upload-adjustments>

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Merchant API — products: <https://developers.google.com/merchant/api/guides/products/add-manage>

Merchant API — data sources: <https://developers.google.com/merchant/api/guides/data-sources/api-sources>

Meta Conversions API: <https://developers.facebook.com/documentation/ads-commerce/conversions-api>

Meta CAPI + server-side GTM/deduplication: <https://developers.facebook.com/documentation/ads-commerce/conversions-api/guides/gtm-server-side>

TikTok Events API deduplication: <https://ads.tiktok.com/help/article/event-deduplication>

TikTok Events API matching: <https://ads.tiktok.com/help/article/how-to-set-up-matching-events-with-events-api>

Snap Conversions API deduplication: <https://developers.snap.com/marketing-api/Conversions-API/Deduplication>

Google Search — ecommerce URL structure: <https://developers.google.com/search/docs/specialty/ecommerce/designing-a-url-structure-for-ecommerce-sites>

Google Search — canonicalization: <https://developers.google.com/search/docs/crawling-indexing/consolidate-duplicate-urls>

Google Search — robots meta directives: <https://developers.google.com/search/docs/crawling-indexing/robots-meta-tag>

Google Search — sitemap construction: <https://developers.google.com/search/docs/crawling-indexing/sitemaps/build-sitemap>

Google Search — pagination & incremental loading:

<https://developers.google.com/search/docs/specialty/ecommerce/pagination-and-incremental-page-loading>

Google Search — managing faceted navigation: <https://developers.google.com/search/docs/crawling-indexing/crawling-managing-faceted-navigation>

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